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Semiconductor devices having air spacer

Abstract

A semiconductor device includes: a stack including a conductive structure and an insulation structure stacked on each other on a substrate, wherein the stack extends in a first direction substantially parallel to an upper surface of the substrate; first, second and third spacers sequentially stacked on each other on a sidewall of the stack in a second direction substantially parallel to the upper surface of the substrate and crossing the first direction; and a capping pattern disposed on the second spacer, wherein: the second spacer is an air spacer including air, and an upper surface of a portion of the third spacer is substantially coplanar with an upper surface of the capping pattern.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-

2021-0150278 filed on Nov. 4, 2021 in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

(2) Example embodiments of the present inventive concept relate to a semiconductor device. More particularly, example embodiments of the present inventive concept relate to a dynamic random-access memory (DRAM) device.

DISCUSSION OF THE RELATED ART

(3) Generally, in a dynamic random-access memory (DRAM) device, a parasitic capacitance may occur between a bit line structure and a contact plug structure adjacent thereto, and thus, an air spacer may be formed adjacent to the bit line structure. For example, a preliminary spacer structure may be formed on the bit line structure, and a conductive structure may be formed on the preliminary spacer structure. In addition, the conductive structure may be patterned to form an opening exposing a portion of the preliminary spacer structure. The preliminary spacer structure exposed by the opening may be partially removed to form an air gap, and an insulation pattern may be formed in the opening so that the air gap may be transformed into an air spacer.

(4) However, the insulation pattern may permeate into the air gap during the formation thereof, and the air spacer might not have the initial volume of the air gap.

SUMMARY

(5) Example embodiments of the present inventive concept provide a semiconductor device having improved characteristics.

(6) According to an example embodiment of the present inventive concept, a semiconductor device includes: a stack including a conductive structure and an insulation structure stacked on each other on a substrate, wherein the stack extends in a first direction substantially parallel to an upper surface of the substrate; first, second and third spacers sequentially stacked on each other on a sidewall of the stack in a second direction substantially parallel to the upper surface of the substrate and crossing the first direction; and a capping pattern disposed on the second spacer, wherein: the second spacer is an air spacer including air, and an upper surface of a portion of the third spacer is substantially coplanar with an upper surface of the capping pattern.

(7) According to an example embodiment of the present inventive concept, a semiconductor device includes: a stack including a conductive structure and an insulation structure stacked on each other on a substrate in a vertical direction substantially perpendicular to an upper surface of the substrate, wherein the stack extends in a first direction substantially parallel to the upper surface of the substrate; first, second and third spacers sequentially stacked on each other on a sidewall of the stack in a second direction substantially parallel to the upper surface of the substrate and crossing the first direction; and a capping pattern covering upper surfaces of the second and third spacers, wherein: the second spacer is an air spacer including air, and the capping pattern includes a material different from that of the third spacer.

(8) According to an example embodiment of the present inventive concept, a semiconductor device includes: an active pattern disposed on a substrate; a gate structure buried in an upper portion of the active pattern, wherein the gate structure extends in a first direction parallel to an upper surface of the substrate; a bit line structure disposed on the active pattern and extending in a second direction parallel to the upper surface of the substrate and substantially perpendicular to the first direction, first, second and third spacers sequentially stacked on each other in the first direction on a sidewall of the bit line structure; a capping pattern disposed on the second spacer; a fourth spacer disposed on the capping pattern and the third spacer, wherein the fourth spacer contacts an outer sidewall of the first spacer; a contact plug structure disposed on an upper surface of each end portion of the active pattern; and a capacitor disposed on the contact plug structure, wherein: the second spacer is an air spacer including air, and an upper surface of a portion of the third spacer, which is adjacent to the capping pattern, is substantially coplanar with an upper surface of the capping pattern.

(9) The semiconductor device according to an example embodiment may include the air spacer

having a sufficient volume, so that the capacitance between neighboring conductive structure, e.g., between the bit line structure and the contact plug structure may decrease, and the contact plug structure including the metal silicide pattern may have enhanced electrical characteristics.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. **1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21** and **22** are plan views and cross-sectional views illustrating a method of manufacturing a semiconductor device according to an example embodiment of the present inventive concept.

(2) FIG. **23** is a cross-sectional view illustrating a semiconductor device according to an example embodiment of the present inventive concept.

(3) FIGS. **24, 25, 26** and **27** are cross-sectional views illustrating a method of manufacturing a semiconductor device according to an example embodiment of the present inventive concept.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(4) Hereinafter, example embodiments of the present inventive concept will be described in detail with reference to the accompanying drawings. It will be understood that, although the terms “first,” “second,” and/or “third” may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms are only used to distinguish one element, component, region, layer or section from another element, component, region, layer or section. Thus, a first element, component, region, layer or section discussed below could be termed a second or third element, component, region, layer or section without departing from the spirit and scope of the present inventive concept.

(5) FIGS. **1** to **22** are plan views and cross-sectional views illustrating a method of manufacturing a semiconductor device according to an example embodiment of the present inventive concept.

Particularly, FIGS. **1, 3, 6, 8, 12** and **20** are the plan views, and each of FIGS. **2, 4-5, 7, 9-11, 13-19** and **21-22** includes cross-sections taken along lines B-B' and C-C' of a corresponding plan view.

(6) Hereinafter, in the specification, two directions substantially parallel to an upper surface of a substrate and substantially perpendicular to each other may be referred to as first and second directions D1 and D2, respectively, and a direction substantially parallel to the upper surface of the substrate and having an acute angle with respect to the first and second directions D1 and D2 may be referred to as a third direction D3.

(7) Referring to FIGS. **1** and **2**, an active pattern **103** may be formed on the substrate **100**, and an isolation pattern **112** may be formed to cover a sidewall of the active pattern **103**.

(8) The substrate **100** may include, for example, silicon, germanium, silicon-germanium, or a III-V group compound semiconductor, such as GaP, GaAs, or GaSb. In an example embodiment of the present inventive concept, the substrate **100** may be a silicon-on-insulator (SOI) substrate or a germanium-on-insulator (GOI) substrate.

(9) The active pattern **103** may be formed by removing an upper portion of the substrate **100** to form a first recess, and may extend in the third direction D3, from a plan view. In an example embodiment of the present inventive concept, a plurality of active patterns **103** may be spaced apart from each other in the first and second directions D1 and D2. The isolation pattern **112** may be formed in the first recess, and may include an oxide, e.g., silicon oxide.

(10) The active pattern **103** and the isolation pattern **112** may be partially removed to form a second recess extending in the first direction D1.

(11) A gate structure **170** may be formed in the second recess. The gate structure **170** may include a gate insulation pattern **120**, a first barrier pattern **130**, a first conductive pattern **140**, a second conductive pattern **150** and a gate mask **150**. The gate insulating pattern **120** may be disposed on a

bottom surface and a sidewall of the second recess. The first barrier pattern **130** may be disposed on a portion of the gate insulation pattern **120** that is disposed on the bottom surface and a lower sidewall of the second recess. The first conductive pattern **140** may be disposed on the first barrier pattern **130** and may fill a lower portion of the second recess. The second conductive pattern **150** may be disposed on the first barrier pattern **130** and an upper surface of the first conductive pattern **140**, and the gate mask **160** may be disposed on an upper surface of the second conductive pattern **150** and an upper inner sidewall of the gate insulation pattern **120** and filling an upper portion of the second recess. The first barrier pattern **130**, the first conductive pattern **140** and the second conductive pattern **150** may form a gate electrode.

(12) The gate insulation pattern **120** may include an oxide, e.g., silicon oxide. The first barrier pattern **130** may include a metal nitride, e.g., titanium nitride, tantalum nitride, etc. The first conductive pattern **140** may include, for example, a metal, a metal nitride, a metal silicide, doped polysilicon, etc. The second conductive pattern **150** may include, for example, doped polysilicon, and the gate mask **160** may include a nitride, e.g., silicon nitride.

(13) In an example embodiment of the present inventive concept, the gate structure **170** may extend in the first direction **D1**, and a plurality of gate structures **170** may be spaced apart from each other in the second direction **D2**.

(14) Referring to FIGS. **3** and **4**, first and second insulation layers **180** and **190** may be formed on the substrate **100**.

(15) The first insulation layer **180** may include an oxide, e.g., silicon oxide, and the second insulation layer **190** may include a nitride, e.g., silicon nitride.

(16) A natural oxide layer including, e.g., silicon oxide may be formed on the second insulation layer **190**, which may be referred to as a third insulation layer **200**. The first to third insulation layers **180**, **190** and **200** may be referred to as an insulation layer structure **210**.

(17) The insulation layer structure **210** may be patterned, and the active pattern **103**, the isolation pattern **112** and the gate mask **160** included in the gate structure **170** may be partially etched using the patterned insulation layer structure **210** as an etching mask to form a first opening **230**. In an example embodiment of the present inventive concept, the insulation layer structure **210** remaining after the etching process may have a shape of a circle or an ellipse in a plan view, and a plurality of insulation layer structures **210** may be spaced apart from each other in the first and second directions **D1** and **D2**. Each of the insulation layer structures **210** may overlap end portions in the third direction **D3** of neighboring ones of the active patterns **103** in a vertical direction substantially perpendicular to an upper surface of the substrate **100**.

(18) Referring to FIG. **5**, a third conductive layer **240**, a second barrier layer **250**, a fourth conductive layer **260** and a first mask layer **270** may be sequentially stacked on the insulation layer structure **210** and the active pattern **103**, the isolation pattern **112** and the gate structure **170**, which are exposed by the first opening **230**. The third conductive layer **240**, the second barrier layer **250** and the fourth conductive layer **260** form a conductive layer structure. The third conductive layer **240** may fill the first opening **230**.

(19) The third conductive layer **240** may include, e.g., doped polysilicon. The second barrier layer **250** may include a metal silicon nitride, e.g., titanium silicon nitride. The fourth conductive layer **260** may include a metal, e.g., tungsten, and the first mask layer **270** may include a nitride, e.g., silicon nitride.

(20) Referring to FIGS. **6** and **7**, a first etch stop layer **360** may be formed on the first mask layer **270**. The first etch stop layer **360** may include a nitride, e.g., silicon nitride.

(21) Referring to FIGS. **8** and **9**, a first capping layer may be formed on the first etch stop layer **360**, and may be patterned to form a first capping pattern **385**.

(22) In an example embodiment of the present inventive concept, the first capping pattern **385** may extend in the second direction **D2**, and a plurality of first capping patterns **385** may be spaced apart from each other in the first direction **D1**. The first capping pattern **385** may include a nitride, e.g.,

silicon nitride.

(23) The first etch stop layer **360**, the first mask layer **270**, the fourth conductive layer **260**, the second barrier layer **250** and the third conductive layer **240** may be sequentially etched using the first capping pattern **385** as an etching mask.

(24) By the etching process, a third conductive pattern **245**, a second barrier pattern **255**, a fourth conductive pattern **265**, a first mask **275**, a first etch stop pattern **365** and the first capping pattern **385** may be sequentially stacked on the first opening **230**, and outside of the first opening **230**, a third insulation pattern **205**, the third conductive pattern **245**, the second barrier pattern **255**, the fourth conductive pattern **265**, the first mask **275**, the first etch stop pattern **365** and the first capping pattern **385** may be sequentially stacked on the second insulation layer **190** of the insulation layer structure **210**.

(25) Hereinafter, the third conductive pattern **245**, the second barrier pattern **255**, the fourth conductive pattern **265**, the first mask **275**, the first etch stop pattern **365** and the first capping pattern **385** that are sequentially stacked may be referred to as a bit line structure **395**. The bit line structure **395** may include a conductive structure including the third conductive pattern **245**, the second barrier pattern **255** and the fourth conductive pattern **265** sequentially stacked on each other. In addition, the bit line structure **395** may include an insulation structure disposed on the conductive structure and including the first mask **275**, the first etch stop pattern **365** and the first capping pattern **385** sequentially stacked on the conductive structure. In an example embodiment of the present inventive concept, the bit line structure **395** may extend in the second direction D2 on the substrate **100**, and a plurality of bit line structures **395** may be spaced apart from each other in the first direction D1.

(26) Referring to FIG. **10**, a first spacer layer may be formed on the substrate **100** having the bit line structure **395** thereon, and fourth and fifth insulation layers may be sequentially formed on the first spacer layer.

(27) The first spacer layer may also cover a sidewall of the third insulation pattern **205** that is disposed under a portion of the bit line structure **395** that is disposed on the second insulation layer **190**, and the fifth insulation layer may fill a remaining portion of the first opening **230**.

(28) The first spacer layer may include a nitride, e.g., silicon nitride. The fourth insulation layer may include an oxide, e.g., silicon oxide, and the fifth insulation layer may include a nitride, e.g., silicon nitride.

(29) The fourth and fifth insulation layers may be etched by an etching process. In an example embodiment of the present inventive concept, the etching process may be performed by a wet etching process using phosphoric acid, SCI and hydrofluoric acid as an etching solution, and other portions of the fourth and fifth insulation layers except for portions of the fourth and fifth insulation layers in the first opening **230** may be removed. Thus, most of a surface of the first spacer layer, that is, other portions of the first spacer layer except for the portion thereof in the first opening **230** may be exposed, and the portions of the fourth and fifth insulation layers remaining in the first opening **230** may form fourth and fifth insulation patterns **410** and **420**, respectively.

(30) A second spacer layer may be formed on the exposed surface of the first spacer layer and the fourth and fifth insulation patterns **410** and **420** in the first opening **230**, and may be anisotropically etched to form a preliminary second spacer **430** on the surface of the first spacer layer and the fourth and fifth insulation patterns **410** and **420** to cover a sidewall of the bit line structure **395**. The preliminary second spacer layer may include an oxide, e.g., silicon oxide.

(31) A dry etching process may be performed using the first capping pattern **385** and the preliminary second spacer **430** as an etching mask to form a second opening **440** exposing an upper surface of the active pattern **103**, and upper surfaces of the isolation pattern **112** and the gate mask **160** may also be exposed by the second opening **440**.

(32) By the dry etching process, a portion of the first spacer layer on the upper surfaces of the first capping pattern **385** and the second insulation layer **190** may be removed, and thus a first spacer

400 may be formed to cover the sidewall of the bit line structure **395**. Additionally, during the dry etching process, the first and second insulation layers **190** and **200** may be partially removed, and first and second insulation patterns **195** and **205** may remain under the bit line structure **395**. The first to third insulation patterns **195**, **205** and **215**, which are sequentially stacked on each other under the bit line structure **395**, may form a first insulation pattern structure.

(33) Referring to FIG. **11**, a third spacer layer may be formed on the upper surface of the first capping pattern **385**, an outer sidewall of the preliminary second spacer **430**, portions of the upper surfaces of the fourth and fifth insulation patterns **410** and **420**, and upper surfaces of the active pattern **103**, which is exposed by the second opening **440**, the isolation pattern **112**, which is exposed by the second opening **440**, and the gate mask **160** that is exposed by the second opening **440**, and may be anisotropically etched to form a third spacer **450** covering the sidewall of the bit line structure **395**. The third spacer layer may include a nitride, e.g., silicon nitride.

(34) The first spacer **400**, the preliminary second spacer **430** and the third spacer **450**, which are sequentially stacked on the sidewall of the bit line structure **395** in a horizontal direction substantially parallel to the upper surface of the substrate **100**, may be referred to as a preliminary spacer structure **460**.

(35) A lower contact plug layer **470** may be formed to fill the second opening **440**, and may be planarized until the upper surface of the first capping pattern **385** is exposed. For example, an upper surface of the lower contact plug layer **470** may be coplanar with the upper surface of the first capping pattern **385**.

(36) In an example embodiment of the present inventive concept, the lower contact plug layer **470** may extend in the second direction **D2**, and a plurality of lower contact plug layers **470** may be spaced apart from each other in the first direction **D1** by the bit line structures **395**. The lower contact plug layer **470** may include, e.g., doped polysilicon.

(37) Referring to FIGS. **12** and **13**, a second mask having a plurality of third openings spaced apart from each other in the second direction **D2**, each of which may extend in the first direction **D1**, may be formed on the first capping pattern **385** and the lower contact plug layer **470**, and the lower contact plug layer **470** may be etched using the second mask as an etching mask.

(38) In an exemplary embodiment of the present inventive concept, each of the third openings may overlap the gate structure **170** in the vertical direction. By the etching process, a fourth opening exposing an upper surface of the gate mask **160** of the first gate structure **170** may be formed between the bit line structures **395**.

(39) After removing the second mask, a second capping pattern **480** may be formed to fill the fourth opening. The second capping pattern **480** may include a nitride, e.g., silicon nitride. In an example embodiment of the present inventive concept, a plurality of second capping patterns **480** may be spaced apart from each other in the second direction **D2** between the bit line structures **395** neighboring in the first direction **D1**. For example, the second capping patterns **480** may extend in the first direction **D1** while the bit line structures **395** extend in the second direction **D2**.

(40) Thus, the lower contact plug layer **470** may be divided into a plurality of lower contact plugs **475** spaced apart from each other in the second direction **D2** by the second capping patterns **480**.

(41) Referring to FIG. **14**, an upper portion of the lower contact plug **475** may be removed to expose an upper portion of the preliminary spacer structure **460** on the sidewall of the bit line structure **395**, and upper portions of the preliminary second spacer **430** and the third spacer **450** of the exposed preliminary spacer structure **460** may be removed.

(42) An upper portion of the lower contact plug **475** may be removed by, e.g., an etch back process, and the upper portions of the preliminary second spacer **430** and the third spacer **450** may be removed by, e.g., a wet etching process.

(43) The preliminary second spacer **430** may be further removed to form an air gap **432**. The preliminary second spacer **430** may be removed by, e.g., a dry etching process. The air gap **432** may be formed between the lower contact plug **475** and the bit line structure **395**.

(44) In an example embodiment of the present inventive concept, not only a portion of the preliminary second spacer **430** under the lower contact plug **475** but also a portion of the preliminary second spacer **430** covered by the second capping pattern **480** may be removed.

(45) As the air gap **432** is formed, an upper portion of the first spacer **400** and upper surfaces of the fourth and fifth insulation patterns **410** and **420** may be exposed.

(46) Referring to FIG. **15**, a first sacrificial pattern **700** may be formed to fill the air gap **432**.

(47) The first sacrificial pattern **700** may be formed by forming a first sacrificial layer on the bit line structure **395**, the first and third spacers **400** and **450**, the lower contact plug **475** and the second capping pattern **480**, and performing an etch back process on the first sacrificial layer until an upper surface of the lower contact plug **475** is exposed. A stripping process may be performed.

(48) Thus, an upper surface of the first sacrificial pattern **700** may be substantially coplanar with upper surfaces of the lower contact plug **475** and the third spacer **450**.

(49) In an example embodiment of the present inventive concept the first sacrificial pattern **700** may include a pyrolysis material that may be decomposed by heat.

(50) Referring to FIG. **16**, an upper portion of the first sacrificial pattern **700** may be removed to form a third recess, and a third capping pattern **715** may be formed in the third recess.

(51) In an example an embodiment of the present inventive concept, a third capping layer **710** (refer to FIG. **24**) may be formed on the bit line structure **395**, the first and third spacers **400** and **450**, the first sacrificial pattern **700**, the lower contact plug **475** and the second capping pattern **480** by, e.g., an ALD process to fill the third recess, and a wet etching process may be performed on the third capping layer so that the third capping pattern **715** may be formed in the third recess. Thus, an upper surface of the third capping pattern **715** may be substantially coplanar with the upper surfaces of the lower contact plug **475** and the third spacer **450**.

(52) In an example an embodiment of the present inventive concept, the third capping pattern **715** may include an oxide, e.g., silicon oxide.

(53) Referring to FIG. **17**, the substrate **100** may be heated to remove the first sacrificial pattern **700**.

(54) In an example embodiment of the present inventive concept, the substrate **100** having the first sacrificial pattern **700** thereon may be heated to a temperature of more than about 150° C., and the first sacrificial pattern **700** including the pyrolysis material may be removed.

(55) Thus, a second spacer **435** including air may be formed between the first and third spacers **400** and **450**, and may be referred to as an air spacer **435**.

(56) The first to third spacers **400**, **435** and **450** sequentially stacked on the sidewall of the bit line structure **395** in the horizontal direction may form a spacer structure **465**.

(57) Referring to FIG. **18**, a fourth spacer layer may be formed on the bit line structure **395**, the spacer structure **465**, the third capping pattern **715**, the lower contact plug **475** and the second capping pattern **480**, and may be, for example, anisotropically etched to form a fourth spacer **490** on a sidewall of a portion of the first spacer **400** on an upper sidewall of the bit line structure **395**.

(58) The fourth spacer **490** formed by the anisotropic etching process may cover an upper surface of the third capping pattern **715** and at least a portion of an upper surface of the third spacer **450**. Thus, an upper portion of the lower contact plug **475** may be partially removed during the anisotropic etching process, and a portion of the third spacer **450** not covered by the fourth spacer **490** may be partially removed.

(59) A fifth spacer layer may be formed on the bit line structure **395**, the first spacer **400**, the fourth spacer **490**, the lower contact plug **475** and the second capping pattern **480**, and may be etched to form a fifth spacer on a sidewall of the fourth spacer **490**. An etching process may be performed using the bit line structure **395**, the first spacer **400**, the fourth spacer **490**, and the second capping pattern **480** as an etching mask so that an upper portion of the lower contact plug **475** may be further removed. Thus, an upper surface of the lower contact plug **475** may be lower than upper surfaces of the second and third spacers **435** and **450**.

(60) A metal silicide pattern **500** may be formed on the lower contact plug **475**. For example, the metal silicide pattern **500** may be formed on the upper surface of lower contact plug **475**. In an example embodiment of the present inventive concept, the metal silicide pattern **500** may be formed by forming a first metal layer on the bit line structure **395**, the first and fourth spacers **400** and **490**, the lower contact plug **475** and the second capping pattern **480**, performing a heat treatment on the first metal layer to perform a silicidation process, and removing an unreacted portion of the first metal layer. In the silicidation process, the first metal layer, which includes a metal, and the lower contact plug **475**, which includes silicon, may react with each other. A height of an upper surface of the metal silicide pattern **500** that may be formed by the silicidation process may slightly increase.

(61) The metal silicide pattern **500** may include, e.g., cobalt silicide, nickel silicide, titanium silicide, etc.

(62) Referring to FIG. **19**, a third barrier layer **530** may be formed on the bit line structure **395**, the first and fourth spacers **400** and **490**, the metal silicide pattern **500** and the second capping pattern **480**, and a second metal layer **540** may be formed on the third barrier layer **530** to fill a space between the bit line structures **395**.

(63) A planarization process may be performed on an upper portion of the second metal layer **540**. The planarization process may include, for example, a CMP process and/or an etch back process.

(64) Referring to FIGS. **20** and **21**, the second metal layer **540** and the third barrier layer **530** may be patterned to form an upper contact plug **549**, and a fifth opening **547** may be formed between the upper contact plugs **549**.

(65) During the formation of the fifth opening **547**, the first and second capping patterns **385** and **480**, the third spacer structure **465**, the fourth spacer **490** and the first mask **275** may be partially removed in addition to the second metal layer **540** and the third barrier layer **530**.

(66) As the fifth opening **547** is formed, the second metal layer **540** and the third barrier layer **530** may be transformed into a second metal pattern **545** and a third barrier pattern **535**, respectively, which may form an upper contact plug **549**. For example, the third barrier pattern **535** may cover a lower surface and a sidewall of the second metal pattern **545**. In an example embodiment of the present inventive concept, a plurality of upper contact plugs **549** may be spaced apart from each other in the first and second directions **D1** and **D2**, and may be arranged in a honeycomb pattern in a plan view. For example, adjacent rows of upper contact plugs **549** may be misaligned with each other. For example, each of the upper contact plugs **103** may overlap an end of an active pattern **103**. Each of the upper contact plugs **549** may have a shape of, for example, a circle, an ellipse, or a polygon.

(67) The lower contact plug **475**, the metal silicide pattern **500** and the upper contact plug **549** sequentially stacked on the substrate **100** may form a contact plug structure.

(68) Referring to FIG. **22**, a sixth insulation pattern **610** may be formed on at least a sidewall of the fifth opening **547**, and a seventh insulation pattern **620** may be formed to fill a remaining portion of the fifth opening **547**.

(69) The sixth insulation pattern **610** may be formed by forming a sixth insulation layer on at least an inner wall of the fifth opening **547**, the upper contact plug **549** and the second capping pattern **480**, and anisotropically etching the sixth insulation layer.

(70) The seventh insulation pattern **620** may be formed by forming a seventh insulation layer on the sixth insulation pattern **610**, the upper contact plug **549** and the second capping pattern **480**, and performing an etch back process on the seventh insulation layer.

(71) Each of the sixth and seventh insulation patterns **610** and **620** may include a nitride, e.g., silicon nitride, and the sixth and seventh insulation patterns **610** and **620** may form a second insulation pattern structure.

(72) A second etch stop layer **630** may be formed on the seventh insulation pattern **620**, the upper contact plug **549** and the second capping pattern **480**, and a mold layer may be formed on the

second etch stop layer **630**. A portion of the mold layer and a portion of the second etch stop layer **630** thereunder may be partially etched to form a sixth opening exposing an upper surface of the upper contact plug **549**.

(73) As the plurality of upper contact plugs **549** is spaced apart from each other in the first and second directions **D1** and **D2**, and may be arranged in a honeycomb pattern or a lattice pattern in a plan view, the sixth openings exposing the upper contact plugs **549** may be arranged with an arrangement corresponding to that of the upper contact plugs **549**. For example, the sixth openings may also be arranged in a honeycomb pattern or a lattice pattern in a plan view.

(74) A lower electrode layer may be formed on a sidewall of the sixth opening, the exposed upper surface of the upper contact plug **549** and the mold layer. A second sacrificial layer may be formed on the lower electrode layer to fill the sixth opening, and the lower electrode layer and the second sacrificial layer may be planarized until an upper surface of the mold layer is exposed to divide the lower electrode layer.

(75) Thus, a lower electrode **640** having a shape of, for example, a cylinder may be formed in the sixth opening. However, if the sixth opening has a small width, the lower electrode **640** may have a shape of, for example, a pillar. The lower electrode **640** may include, e.g., a metal, a metal nitride, a metal silicide, doped polysilicon, etc.

(76) The second sacrificial layer and the mold layer may be removed by, e.g., a wet etching process using, e.g., LAL solution.

(77) A dielectric layer **650** may be formed on a surface of the lower electrode **640** and the second etch stop layer **630**. The dielectric layer **650** may include, e.g., a metal oxide.

(78) An upper electrode **660** may be formed on the dielectric layer **650**. The upper electrode **660** may include, e.g., a metal, a metal nitride, a metal silicide, doped silicon-germanium, etc. In an example embodiment of the present inventive concept, the upper electrode **660** may have a first upper electrode, which includes a metal or a metal nitride and a second upper electrode, which includes doped silicon-germanium.

(79) The lower electrode **640**, the dielectric layer **650** and the upper electrode **660** may form a capacitor **670**.

(80) Upper wirings may be further formed on the capacitor **670** to complete the fabrication of the semiconductor device.

(81) As illustrated above, upper portions of the lower contact plug **475** on the sidewall of the bit line structure **395**, the preliminary second spacer **430** of the preliminary spacer structure **460**, and the third spacer **450** of the preliminary spacer structure **460** may be removed. The preliminary second spacer **430** may be removed to form an air gap **432**, and the first sacrificial pattern **700** including a pyrolysis material may be formed in the air gap **432**. The third capping pattern **715** may be formed to cover an upper surface of the first sacrificial pattern **700**, and the substrate **100** may be heated to remove the first sacrificial pattern **700**, so that the air spacer **435** including air may be formed.

(82) The fourth spacer **490** may be formed to cover the upper surfaces of the third capping pattern **715** and the third spacer **450**, and the metal silicide pattern **500** may be formed on the lower contact plug **475**. The upper contact plug layer may be formed on the bit line structure **395**, the fourth spacer **490** and the metal silicide pattern **500**, and may be patterned to form the upper contact plug **549**, and the second insulation pattern structure including the sixth and seventh insulation patterns **610** and **620** may be formed in the fifth opening **547** that may be formed during the patterning the upper contact plug layer.

(83) Thus, the air spacer **435** may be formed by heating the first sacrificial pattern **700**, which includes a pyrolysis material, with a top end of the air gap **432**, which may be formed by the first and third spacers **400** and **450**, that is closed by the third capping pattern **715**, and thus, the air gap **432** might not be filled with other materials so that the air spacer **435** may have the initial volume of the air gap **432**.

(84) For example, if the upper contact plug layer is patterned to form the fifth opening **547** exposing the preliminary spacer structure **460**, if the preliminary second spacer **430** exposed by the fifth opening **547** is removed to form an air gap, and if the fifth opening **547** is filled with the second insulation pattern structure so that the air gap is transformed into the air spacer **435**, then the second insulation pattern structure may partially permeate into the air gap, so that the air spacer **435** might not have the initial volume of the air gap.

(85) Additionally, the air spacer **435** might not be formed by exposing the preliminary second spacer **430** through the fifth opening **547** and removing the preliminary second spacer **430**, but may be formed before forming the fifth opening **547**. Thus, the fifth opening **547** might not be formed to a deep depth. If the fifth opening **547** is formed to a deep depth, for example, the metal silicide pattern **500** may be exposed so that a metal included in the metal silicide pattern **500** may partially leak. Thus, in an example embodiment of the present inventive concept, the leakage of the metal included in the metal silicide pattern **500** may be prevented.

(86) As a result, the semiconductor device may include the air spacer **435** having a sufficient volume, so that the parasitic capacitance between neighboring conductive structures, e.g., between the bit line structure **395** and the contact plug structure may decrease, and the contact plug structure including the metal silicide pattern **500** may have enhanced electrical characteristics.

(87) The semiconductor device manufactured by the above processes may have following structural characteristics.

(88) Referring to FIGS. **20** and **22**, the semiconductor device may include the active pattern **103**, the first gate structure **170**, the bit line structure **395**, the spacer structure **465**, the third capping pattern **715**, the fourth spacer **490**, the contact plug structure, and the capacitor **670**. The active pattern **103** may extend in the third direction **D3** on the substrate **100**. The first gate structure **170** may extend in the first direction **D1** in an upper portion of the active pattern **103**. The bit line structure **395** may contact a central upper surface of the active pattern **103** and may extend in the second direction **D2**. The spacer structure **465** may include the first to third spacers **400**, **435** and **450** sequentially stacked in the first direction **D1** on the sidewall of the bit line structure **395**. The third capping pattern **715** may be disposed on the second spacer **435**. The fourth spacer **490** may be disposed on the third capping pattern **715** and the third spacer **450** and may contact the outer sidewall of the first spacer **400**. The contact plug structure may contact each of opposite ends of the active pattern **103**, and the capacitor **670** may be disposed on the contact plug structure.

(89) In an example embodiment of the present inventive concept, the second spacer **435** may be an air spacer, and an upper surface of a portion of the third spacer **450** adjacent to the third capping pattern **715** may be substantially coplanar with an upper surface of the third capping pattern **715**.

(90) In an example embodiment of the present inventive concept, the bit line structure **395** may include the conductive structure and the insulation structure sequentially stacked in the vertical direction. The conductive structure may include the third conductive pattern **245**, the second barrier pattern **255** and the fourth conductive pattern **265** sequentially stacked on each other, and the insulation structure may include the first mask **275**, the first etch stop pattern **365** and the first capping pattern **385** sequentially stacked in the vertical direction on each other.

(91) In an example embodiment of the present inventive concept, the first spacer **400** may cover the sidewall of the bit line structure **395** in the first direction **D1**. For example, the first spacer **400** may entirely cover the sidewall of the bit line structure **395**. The second spacer **435** may overlap along the second direction **D2** a central portion of the sidewall of the bit line structure **395**. For example, the second spacer **435** may vertically extend along the sidewall of the bit line structure **395**.

(92) In an example embodiment of the present inventive concept, the fourth and fifth insulation patterns **410** and **420** including silicon oxide and silicon nitride, respectively, may be stacked on each other between the first and second spacers **400** and **435**.

(93) In an example embodiment of the present inventive concept, the contact plug structure may be formed on the substrate **100** adjacent to the bit line structure **395** in the second direction **D2**, and

may include the lower contact plug **475**, the metal silicide pattern **500** and the upper contact plug **549** sequentially stacked on each other in the vertical direction. The lower contact plug **475** may include, e.g., doped polysilicon, and the upper contact plug **549** may have the first metal pattern **545**, which includes, e.g., a metal, and the third barrier pattern **535**, which includes, e.g., a metal nitride.

(94) FIG. **23** is a cross-sectional view illustrating a semiconductor device according to an example embodiment of the present inventive concept. This semiconductor device may be substantially the same as or similar to that of FIGS. **21** and **22**, except for the position of the metal silicide pattern **500**.

(95) In an example embodiment of the preset inventive concept, an upper surface of the metal silicide pattern **500** may be substantially coplanar with or higher than that of the third capping pattern **715**.

(96) As illustrated above, the fifth opening **547** that may be formed by patterning the upper contact plug layer might not have a deep depth to expose the upper surface of the preliminary second spacer **430** and form the air spacer **435**, and thus, the metal silicide pattern **500** might not be exposed by the fifth opening **547**. Accordingly, the processes illustrated with reference to FIG. **18**, that is, removing the upper portion of the lower contact plug **475** after forming the fifth spacer so as to lower the upper surface of the metal silicide pattern **500** is not needed, and thus, the upper surface of the lower contact plug **475** may have height substantially equal or similar to those of the third spacer **450** and the third capping pattern **715**.

(97) The upper surface of the metal silicide pattern **500** that is disposed on the lower contact plug **475** may be slightly higher than the upper surface of the lower contact plug **475**, and thus, the upper surface of the metal silicide pattern **500** may be substantially coplanar with or slightly higher than those of the third capping pattern **715** and the third spacer **450**.

(98) FIGS. **24** to **27** are cross-sectional views illustrating a method of manufacturing a semiconductor device according to an example embodiment of the present inventive concept, and each of FIGS. **24** to **27** includes cross-sections taken along lines B-B' and C-C' of a corresponding plan view. This method may include processes substantially the same as or similar to those illustrated with reference to FIGS. **1** to **22**, and repeated explanations thereof may be omitted herein.

(99) Referring to FIG. **24**, processes substantially the same as or similar to those illustrated with reference to FIGS. **1** to **15** may be performed, and the third capping layer **710** may be formed by, e.g., an ALD process on the bit line structure **395**, the first and third spacers **400** and **450**, the first sacrificial pattern **700**, the lower contact plug **475** and the second capping pattern **480**.

(100) The substrate **100** may be heated to remove the first sacrificial pattern **700** so that the second spacer **435** including air may be formed between the first and third spacers **400** and **450**.

(101) Referring to FIG. **25**, a third sacrificial layer may be formed on the third capping layer **710**, and an etch back process may be performed on the third sacrificial layer to remove an upper portion thereof, so that a third sacrificial pattern **720** may be formed.

(102) A portion of the third capping layer **710** exposed by the third sacrificial pattern **720** may be removed by, e.g., a wet etching process so that a fourth capping **712**, which has an upper surface substantially coplanar with the upper surface of the third sacrificial pattern **720**, may be formed.

(103) The third sacrificial pattern **720** may include, e.g., spin-on-hardmask (SOH), amorphous carbon layer (ACL), etc.

(104) Referring to FIG. **26**, the third sacrificial pattern **720** may be removed by, e.g., an ashing process and/or a stripping process to expose the fourth capping pattern **712**.

(105) Processes substantially the same as or similar to those illustrated with reference to FIG. **18** may be performed. For example, the fourth spacer layer may be formed on the bit line structure **395**, the fourth capping pattern **712** and the second capping pattern **480**, and may be anisotropically etched to form the fourth spacer **490** on a portion of the first spacer **400** that is on an upper sidewall

of the bit line structure **395**.

(106) Not only the portion of the fourth spacer layer that is on an upper surface of the lower contact plug **475** but also a portion of the fourth spacer layer under the lower contact plug **475** may be removed to expose the upper surface of the lower contact plug **475**, and thus, a fifth capping pattern **714** covered by the fourth spacer **490** may be formed on the second and third spacers **435** and **450**.

(107) The metal silicide pattern **500** may be formed on the upper surface of the lower contact plug **475**, and as mentioned above, in an example embodiment of the present inventive concept, the fifth spacer may be formed on the sidewall of the fourth spacer **490** before forming the metal silicide pattern **500**.

(108) Referring to FIG. **27**, processes substantially the same as or similar to those illustrated with reference to FIGS. **19** to **22** may be performed to complete the fabrication of the semiconductor device.

(109) The semiconductor device manufactured by the above processes may include the fifth capping pattern **714** covering the upper surfaces of the second and third spacers **435** and **450**. The upper surface of the metal silicide pattern **500** may be substantially coplanar with or slightly higher than the upper surfaces of portions of the second and third spacers **435** and **450** adjacent to the metal silicide pattern **500** in the first direction **D1**, and a lower surface of the metal silicide pattern **500** may be lower than the upper surfaces of the portions of the second and third spacers **435** and **450**.

(110) In an example embodiment of the present inventive concept, the fifth capping pattern **714** may have a cross-section in the second direction **D2** having a bent shape, e.g., an “L” shape.

(111) While the present inventive concepts have been described with reference to example embodiments thereof, it will be understood by those of ordinary skill in the art that various changes in form and details may be made thereto without departing from the spirit and scope of the present inventive concept.

Claims

1. A semiconductor device comprising: a stack including a conductive structure and an insulation structure stacked on each other on a substrate, wherein the stack extends in a first direction substantially parallel to an upper surface of the substrate; first, second and third spacers sequentially stacked on each other on a sidewall of the stack in a second direction substantially parallel to the upper surface of the substrate and crossing the first direction; and a capping pattern disposed on the second spacer, wherein: the second spacer is an air spacer including air, and an upper surface of a portion of the third spacer is substantially coplanar with an upper surface of the capping pattern.
2. The semiconductor device according to claim 1, wherein the first and third spacers include a nitride, and the capping pattern includes an oxide.
3. The semiconductor device according to claim 1, further comprising a fourth spacer disposed on the capping pattern and the third spacer, wherein the fourth spacer contacts an outer sidewall of the first spacer.
4. The semiconductor device according to claim 1, wherein the conductive structure includes first, second and third conductive patterns sequentially stacked in a vertical direction with respect to the upper surface of the substrate, wherein the first conductive pattern includes doped polysilicon, wherein the second conductive pattern includes a metal nitride, and wherein the third conductive pattern includes a metal.
5. The semiconductor device according to claim 1, wherein the insulation structure includes first, second and third insulation patterns sequentially stacked in a vertical direction with respect to the upper surface of the substrate, and wherein each of the first to third insulation patterns includes silicon nitride.

6. The semiconductor device according to claim 1, wherein the first spacer entirely covers the sidewall of the stack in the second direction, and the second spacer overlaps in the second direction a central sidewall of the stack.
 7. The semiconductor device according to claim 6, further comprising first and second insulation patterns disposed between the first spacer and the second spacer, wherein the first insulation pattern includes silicon oxide, and the second insulation pattern includes silicon nitride.
 8. The semiconductor device according to claim 1, further comprising a contact plug structure disposed on a portion of the substrate adjacent to the stack in the second direction, wherein the contact plug structure includes a lower contact plug, a metal silicide pattern and an upper contact plug sequentially stacked on each other in a vertical direction with respect to the upper surface of the substrate.
 9. The semiconductor device according to claim 8, wherein an upper surface of the metal silicide pattern is substantially coplanar with or higher than an upper surface of the capping pattern.
 10. A semiconductor device comprising: an active pattern disposed on a substrate; a gate structure buried in an upper portion of the active pattern, wherein the gate structure extends in a first direction parallel to an upper surface of the substrate; a bit line structure disposed on the active pattern and extending in a second direction parallel to the upper surface of the substrate and substantially perpendicular to the first direction; first, second and third spacers sequentially stacked on each other in the first direction on a sidewall of the bit line structure; a capping pattern disposed on the second spacer; a fourth spacer disposed on the capping pattern and the third spacer, wherein the fourth spacer contacts an outer sidewall of the first spacer; a contact plug structure disposed on an upper surface of each end portion of the active pattern; and a capacitor disposed on the contact plug structure, wherein: the second spacer is an air spacer including air, and an upper surface of a portion of the third spacer, which is adjacent to the capping pattern, is substantially coplanar with an upper surface of the capping pattern.
 11. The semiconductor device according to claim 10, wherein the contact plug structure includes a lower contact plug, a metal silicide pattern and an upper contact plug sequentially stacked on each other, and wherein an upper surface of the metal silicide pattern is substantially coplanar with or higher than the upper surface of the capping pattern.
 12. The semiconductor device according to claim 10, wherein the bit line structure includes a conductive structure and an insulation structure stacked on each other on the substrate, and wherein the first spacer entirely covers the sidewall in the second direction, and the second spacer overlaps in the second direction a central sidewall of the bit line structure.
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